



SECTION 1: GPO ANATOMY *(Group Policy Object Deep Dive)*

WHAT IS A GPO?

A Group Policy Object (GPO) is a collection of settings that you manage centrally and apply to users and computers in your Active Directory environment.



Centralized management

Configure many settings from one place.



Consistent & secure

Enforce standards across your organization.



Automatic application

Settings are applied based on scope and processing rules.

THE TWO MAIN BRANCHES OF A GPO



COMPUTER CONFIGURATION

Settings that apply to the computer itself, regardless of who is logged on.

POLICIES



Software Settings

Assign or publish software (e.g., deploy HR software).



Windows Settings

Configure system settings (e.g., Configure Windows Firewall, audit policy).



Administrative Templates

Policy-based configuration using ADMX templates (e.g., Disable USB storage).



PREFERENCES



Windows Settings

Configure specific Windows settings using preference items (e.g., map a drive, set registry values).



Control Panel Settings

Configure Control Panel items (e.g., set power options, printers).



USER CONFIGURATION

Settings that apply to the user, wherever they log on—on any computer.

POLICIES



Software Settings

Install or publish applications for the user (e.g., deploy Visio to users).



Windows Settings

Configure user-related Windows settings (e.g., folder redirection).



Administrative Templates

User configuration policies (e.g., hide Control Panel items, set Chrome settings).



PREFERENCES



Windows Settings

Configure user-specific settings (e.g., map a network drive for the user, set registry values).



Control Panel Settings

Configure user Control Panel items (e.g., desktop background, email settings).

GPO STRUCTURE (VISUAL TREE)



GPO
(Group Policy Object)



Computer
Configuration



User
Configuration



Policies

- Software Settings
- Windows Settings
- Administrative Templates



Preferences

- Windows Settings
- Control Panel Settings



Policies

- Software Settings
- Windows Settings
- Administrative Templates



Preferences

- Windows Settings
- Control Panel Settings

COMPUTER CONFIGURATION vs USER CONFIGURATION

	 COMPUTER CONFIGURATION	 USER CONFIGURATION
 Applies to	The computer object in AD	The user object in AD
 When applied	During computer startup and user logon (before the user environment is available)	During user logon (after the user environment is available)
 Follows the user?	No – stays with the computer	Yes – follows the user to any computer
 Typical use cases	• Configure Windows Firewall • Deploy HR software to all PCs • Enforce BitLocker encryption • Manage device security settings	• Map user drives • Redirect Documents folder • Set desktop background • Deploy user applications



BEGINNER TIP: Think of Policies as rules you must enforce and Preferences as settings you prefer but can be more flexible.

SECTION 2: ACTIVE DIRECTORY & GPO ARCHITECTURE (ABC COMPANY EXAMPLE)

This section illustrates the Active Directory structure and Group Policy Object (GPO) strategy for ABC Company (domain: abc.local).

ABC COMPANY OVERVIEW

Domain: abc.local

Domain Functional Level: Windows Server 2016

Forest Functional Level: Windows Server 2016

Headquarters: New York, USA

Departments: HR, Finance, IT, Sales

ACTIVE DIRECTORY STRUCTURE

abc.local (Forest Root / Domain Root)

Users

HR

Finance

IT

Sales

Computers

Workstations

Servers

Groups

Global Groups (GG-*)

GG-HR-Users

GG-Finance-Users

GG-IT-Users

GG-Sales-Users

Domain Local Groups (DL-*)

DL-File-HR

DL-File-Finance

DL-File-IT

DL-File-Sales

DL-Print-HR

DL-Remote-Access

DL-Helpdesk

GROUPS STRATEGY

GLOBAL GROUPS (GG)

Used for permissions and role-based access.

GG-HR-Users

GG-Finance-Users

GG-IT-Users

GG-Sales-Users

DOMAIN LOCAL GROUPS (DL)

Used to assign permissions on resources (files, printers, etc.).

DL-File-HR

DL-File-Finance

DL-File-IT

DL-File-Sales

DL-Print-HR

DL-Remote-Access

DL-Helpdesk

GROUP POLICY OBJECTS BY DEPARTMENT				
DEPARTMENT / SCOPE	GPO NAME	SCOPE (LINKED OU)	KEY SETTINGS (SUMMARY)	APPLIES TO
HR	GPO-HR-Baseline	_Users\HR	Desktop settings, Start Menu, Power Management, Folder Redirection (Documents, Desktop)	All HR Users
	GPO-HR-Drive-Mapping	_Users\HR	Map H: (\\fileserver\HR\$\Home), M: (\\fileserver\HR\$\Dept)	HR Users
	GPO-HR-Software	_Users\HR	Deploy: Adobe Reader, 7-Zip, Microsoft Office (365 Apps)	HR Users
	GPO-HR-Restrictions	_Users\HR	Disable Control Panel items, Restrict USB storage, Block access to CMD	HR Users
	GPO-HR-Printing	_Users\HR	Deploy HR Default Printer (\\printserver\HR-PRN1)	HR Users
	GPO-HR-Security	_Users\HR	Password policy (length, complexity), Screen lock (10 min)	HR Users
	GPO-HR-Logon-Scripts	_Users\HR	Run logon script: \\fileserver\scripts\hr\logon.bat	HR Users
FINANCE	GPO-Finance-Baseline	_Users\Finance	Desktop settings, Folder Redirection, Power Management	Finance Users
	GPO-Finance-Drive-Mapping	_Users\Finance	Map H: (\\fileserver\Finance\$\Home), M: (\\fileserver\Finance\$\Dept)	Finance Users
	GPO-Finance-Software	_Users\Finance	Deploy: Adobe Reader, 7-Zip, Excel Add-ins	Finance Users
	GPO-Finance-Restrictions	_Users\Finance	Restrict Registry Tools, Block Install of unapproved software	Finance Users
	GPO-Finance-Printing	_Users\Finance	Deploy Finance Printer (\\printserver\FIN-PRN1)	Finance Users
	GPO-Finance-Security	_Users\Finance	Password policy, Screen lock (10 min), Encryption (EFS)	Finance Users
	GPO-Finance-Logon-Scripts	_Users\Finance	Run logon script: \\fileserver\scripts\finance\logon.bat	Finance Users
IT	GPO-IT-Baseline	_Users\IT	Desktop settings for IT, Enable Dev Tools, Power mgmt	IT Users
	GPO-IT-Drive-Mapping	_Users\IT	Map H: (\\fileserver\IT\$\Home), M: (\\fileserver\IT\$\Dept), T: (\\tools)	IT Users
	GPO-IT-Software	_Users\IT	Deploy: Sysinternals, VS Code, Chrome, WinSCP	IT Users
	GPO-IT-Restrictions	_Users\IT	Allow access to Admin tools, Disable User Account Control prompts	IT Users
	GPO-IT-Printing	_Users\IT	Deploy IT Printer (\\printserver\IT-PRN1)	IT Users
	GPO-IT-Security	_Users\IT	Advanced audit policy, BitLocker (if supported)	IT Users
	GPO-IT-Logon-Scripts	_Users\IT	Run logon script: \\fileserver\scripts\it\logon.bat	IT Users
SALES	GPO-Sales-Baseline	_Users\Sales	Desktop settings, Folder Redirection, Power Management	Sales Users
	GPO-Sales-Drive-Mapping	_Users\Sales	Map H: (\\fileserver\Sales\$\Home), M: (\\fileserver\Sales\$\Dept)	Sales Users
	GPO-Sales-Software	_Users\Sales	Deploy: Adobe Reader, Chrome, Teams	Sales Users
	GPO-Sales-Restrictions	_Users\Sales	Disable Cmd, Restrict USB storage, Block Control Panel	Sales Users
	GPO-Sales-Printing	_Users\Sales	Deploy Sales Printer (\\printserver\SALES-PRN1)	Sales Users
	GPO-Sales-Security	_Users\Sales	Password policy, Screen lock (10 min)	Sales Users
	GPO-Sales-Logon-Scripts	_Users\Sales	Run logon script: \\fileserver\scripts\sales\logon.bat	Sales Users
COMPANY WIDE	GPO-Company-Baseline	abc.local (Domain Root)	Enforce password policy, audit policy, NTP settings, Disable Guest account	All Users & Computers
	GPO-Company-Logon-Banner	abc.local (Domain Root)	Legal logon banner, consent text	All Users
	GPO-Company-Security	abc.local (Domain Root)	Firewall (Domain profile), Defender settings, UAC	All Computers
	GPO-Company-Updates	abc.local (Domain Root)	Windows Update for Business – approved updates	All Computers

SERVER GPOS (APPLIED TO _COMPUTERS\SERVERS)			
GPO NAME	SCOPE (OU)	KEY SETTINGS	APPLIES TO
GPO-DC-Security	_Computers\Servers	Enforce password security hardening, Kerberos policy, NTDS settings, Audit policy	Domain Controllers
GPO-FileServer-Policy	_Computers\Servers	SMB settings, Folder Redirection targets, Disk quotas, Shadow Copy	File Servers
GPO-PrintServer-Policy	_Computers\Servers	Printer sharing, Driver deployment, Auditing	Print Servers
GPO-Server-Baseline	_Computers\Servers	Power options, Windows Update, Event logging, NTP	All Servers

WHY SERVER GPOS DIFFER FROM WORKSTATION GPOS

Shield

Servers provide services and must be hardened for stability, security, and performance.

Gear

Focus on role-specific configurations (e.g., AD DS, File, Print, DNS, DHCP).

Document

Do not apply user experience settings (wallpaper, drive mapping, UI restrictions) that are relevant to users.

Bar chart

Different update strategies and reboot management (compared to workstations).

Lock

Ensure availability, auditing, and backup readiness.

GPO Precedence: Local > Site > Domain > OU (nearest parent takes precedence) | Always test GPOs in a staging OU before production rollout.

SECTION 3

DISTRIBUTION GROUP vs SECURITY GROUP



Core difference: **Distribution Groups** are for email only.
Security Groups are for access control (permissions) and can also be used for email.



DISTRIBUTION GROUP (Mailing List)



SECURITY GROUP (Access Control)

	DISTRIBUTION GROUP (Mailing List)	SECURITY GROUP (Access Control)
 Purpose	Send email to multiple people at once (mailing list).	Control access to resources (files, folders, printers, applications).
 Has SID?	 NO Distribution Groups have NO Security Identifier (SID).	 YES Security Groups have a Security Identifier (SID).
 Can grant permissions?	 NO Cannot be granted permissions or access to resources.	 YES Can be granted permissions and access to resources.
 Can be used for email?	 YES Primary purpose is email (mailing list).	 YES (Optional) Can be used for email if needed, but not required.
 Real-world example	Marketing@company.com (All marketing team members receive announcements)	Finance_Dept (Security Group) Granted access to: \\FileServer\Finance (folder)



RULE OF THUMB

If you need to control access to something, you **MUST** use a **Security Group** — **Distribution Groups** are for email only.





SECTION 4: SECURITY GROUP SCOPES — Global / Domain Local / Universal



In Active Directory, a group's **SCOPE** determines (1) **what can be a member** of the group and (2) **where the group can be used** (granted permissions).

SCOPE TYPE	 GLOBAL GROUP (GG)	 DOMAIN LOCAL GROUP (DL)	 UNIVERSAL GROUP (UG)
 1. WHAT MEMBERS IT CAN CONTAIN	Users and groups from the SAME DOMAIN ONLY. ✓ Can contain users ✓ Can contain other Global groups from the same domain	Users and groups from ANY DOMAIN in the forest. ✓ Can contain users from any domain ✓ Can contain Global, Domain Local or Universal groups from any domain	Users and groups from ANYWHERE in the forest. ✓ Can contain users from any domain ✓ Can contain Global, Domain Local or Universal groups from any domain
 2. WHERE IT CAN BE GRANTED PERMISSIONS ("USED ON")	Resources (objects) in the SAME DOMAIN ONLY. Example: File server, printer, OU, share in abc.local domain	Resources in its OWN DOMAIN ONLY. Example: Can be granted permissions only on resources in abc.local domain (even if it has members from other domains)	Resources ANYWHERE in the forest. Example: Can be granted permissions on resources in any domain in the entire forest
 3. REPLICATION SCOPE	Replicates ONLY within its OWN DOMAIN. Stored in domain naming context of that domain.	Replicates ONLY within its OWN DOMAIN. Stored in domain naming context of that domain.	Replicates to the GLOBAL CATALOG in EVERY DOMAIN in the forest. Stored in the forest-wide Global Catalog.
 KEY POINT (SUMMARY)	Same domain in, same domain out.	Anyone in, own domain out.	Anywhere in, anywhere out.

WHEN TO USE GLOBAL GROUP (GG)

- ✓ Use to represent a role, department or job function **within a single domain**.
- ✓ Assign users to Global groups first.
- ✓ Then add the Global group to a Domain Local group (if access is needed to domain resources).
- ✓ Helps with administration and permissions management.

EXAMPLE



GG-HR-Users

Contains all HR employees from abc.local domain.

WHEN TO USE DOMAIN LOCAL GROUP (DL)

- ✓ Use when you need to grant permissions to **resources in a domain**.
- ✓ Add **Global groups** (or users from other domains) to the **Domain Local group**.
- ✓ Assign this Domain Local group permissions to resources in its **OWN** domain.

EXAMPLE



DL-FileAccess-ABC

Has members from many domains. Used to grant access to a file share in abc.local domain.

WHEN TO USE UNIVERSAL GROUP (UG)

- ✓ Use in multi-domain scenarios where the same group needs access to resources in **multiple domains**.
- ✓ Useful for enterprise-wide groups (e.g., **All Finance Users**).
- ✓ Be mindful: Universal groups replicate more **widely** (to Global Catalog).

EXAMPLE



UG-Finance-All-Domains

Contains finance users from all domains. Can be granted permissions to resources in any domain in the forest.

MEMORY TRICK



- G** **GLOBAL = GO LOCAL**
Stays in its own domain (both membership & use).
- D** **DOMAIN LOCAL = DO IT HERE**
Members from anywhere, but used in **THIS** domain.
- U** **UNIVERSAL = USE ANYWHERE**
Members from anywhere, used anywhere in the forest.

Think: **G** = Group in one domain
D = Do (use) in this domain
U = Use everywhere



BEST PRACTICE MODEL (RECOMMENDED): Users → Global Group (role) → Domain Local Group (resource access) → Permissions on resources

SECTION 5: AGDLP (and AGUDLP) – Microsoft's Recommended Best Practice

A proven, scalable way to manage access and permissions in Active Directory environments.

A = Accounts
(actual user accounts)



Individual users who need access.

G = Global Group
(group accounts by role/department)



Users are grouped based on role or department.

DL = Domain Local Group
(group permissions on a specific resource)



Group that is used to assign permissions on resources.

P = Permission
(the actual access granted on the resource)



The specific access level granted on the resource.

AGDLP IN ACTION – FULL PRACTICAL EXAMPLE

1 ACCOUNTS (A)



Ahmed.HR

Mona.HR

HR employees (user accounts) exist in Active Directory.



2 GLOBAL GROUP (G)



GG-HR-Users
(Global Group)

Ahmed.HR and Mona.HR are added as members of the Global Group "GG-HR-Users".



3 DOMAIN LOCAL GROUP (DL)



DL-File-HR
(Domain Local Group)

"GG-HR-Users" is added as a member of the Domain Local Group "DL-File-HR".



4 PERMISSION (P)



\\FILE01\HR

"DL-File-HR" is granted **Modify Permission** on \\FILE01\HR



Result: Ahmed.HR and Mona.HR can Modify files and folders on \\FILE01\HR through group membership.

WHY AGDLP IS BETTER THAN ASSIGNING PERMISSIONS DIRECTLY TO USER ACCOUNTS



Separation of Concerns

User accounts are managed separately from permissions. IT can manage access by managing groups, not individual users.



Easier Maintainability & Scalability

- Add or remove users by updating group membership.
- Permissions are changed in one place (the DL group), not on every user.
- Reduces risk of permission sprawl and inconsistent access.
- Scales easily as teams and resources grow.



Key Takeaway: Manage people in Global Groups. Manage access in Domain Local Groups. Change once, effective for many.

AGUDLP – FOR MULTI-FOREST ENVIRONMENTS

When users and resources are in different forests, insert a Universal Group (U) between G and DL.



Accounts
(Users)



Global Group
(by role/department)



Universal Group
(across forests)



Domain Local Group
(on resource)



Permission
(on resource)



Universal Groups can be used in any forest in the same Microsoft Entra ID (Azure AD) tenant. Perfect for resource forests, hub-and-spoke, or multi-domain/forest designs.



REAL-WORLD EXAMPLE SUMMARY

Ahmed.HR, Mona.HR (Accounts) →

GG-HR-Users
(Global Group) →



DL-File-HR
(Domain Local Group) →



Modify Permission
on \\FILE01\HR



Secure. Scalable.
Easy to Manage.



SECTION 6: LSDOU — Group Policy Processing Order

Group Policy Objects (GPOs) are processed in a specific order. Understanding this order is essential because later settings can override earlier ones.

LSDOU BREAKDOWN

L = Local Policy
Policy configured on the individual local machine.

S = Site
GPOs linked to the Active Directory Site.

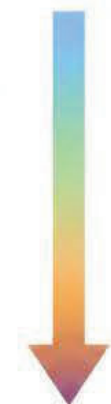
D = Domain
GPOs linked to the domain root.

OU = Organizational Unit
GPOs linked to specific OUs, including nested/child OUs.

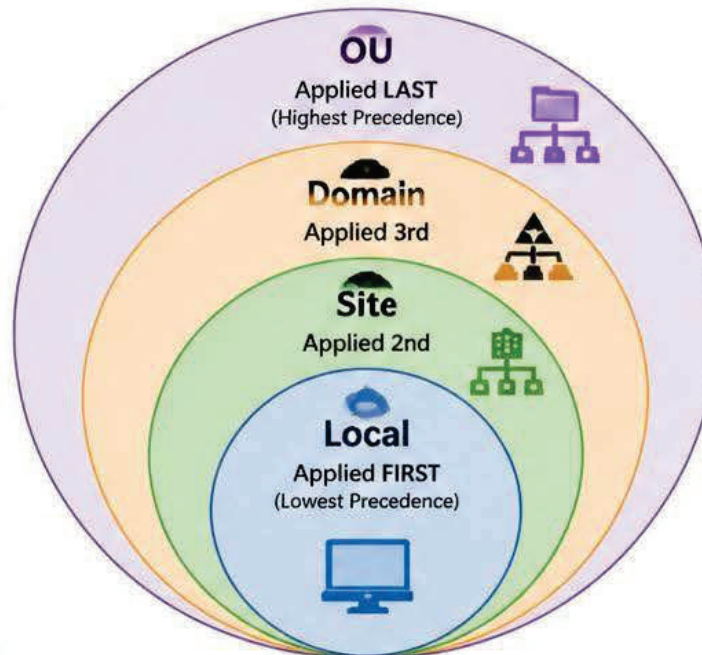
GPOs ARE PROCESSED IN THIS EXACT ORDER:

Local → **Site** → **Domain** → **OU**

Applied
FIRST
(Lowest
Precedence)



Applied
LAST
(Highest
Precedence)



KEY PRECEDENCE RULE

Nested/child OUs are processed **AFTER** their parent OUs.

The OU closest to the actual user/computer object has the **LAST** word and therefore the **HIGHEST** precedence in case of conflicting settings.



Later layers override earlier layers.



If multiple GPOs set the same policy, the setting from the layer processed later wins.

PRACTICAL EXAMPLE: CHANGING THE DESKTOP WALLPAPER

A setting is configured at each level of the LSDOU chain. Watch how the final result is determined.

1. LOCAL (Applied First)



Wallpaper = Blue.jpg
Set on the local machine.

Current value: Blue.jpg



2. SITE (Applied Second)



Wallpaper = Green.jpg
Set by GPO linked to the AD Site.

Overrides previous → Green.jpg



3. DOMAIN (Applied Third)



Wallpaper = Orange.jpg
Set by GPO linked to the domain root.

Overrides previous → Orange.jpg



4. OU (Applied Last)



Wallpaper = Purple.jpg
Set by GPO linked to a child OU (closest OU to the object).

Overrides previous → **PURPLE.jpg**
FINAL RESULT (Highest Precedence)



Because OU is processed last, its setting wins. The user/computer sees Purple.jpg as the final wallpaper.



SECTION 7: GPO LINKING

Link once. Apply everywhere. Control precedence with Link Order.



A single GPO can be LINKED to multiple OUs, Domains, or Sites – no duplicate copies required.

WHAT IS GPO LINKING?

A GPO is linked to a container (OU, Domain, or Site). All objects within that container inherit the GPO. You can link the **SAME GPO** to many containers from a single place.

LINK AN EXISTING GPO vs CREATE A NEW GPO HERE



Link an Existing GPO

- Choose from GPOs already created in this domain.
- Best when the same settings are needed in multiple locations.
- Promotes consistency and easier maintenance.
- Changes to the GPO apply everywhere it is linked.

VS



Create a GPO in this domain and Link it here

- Creates a new GPO and links it to the selected container.
- Best when settings are unique to this location.
- Gives you a dedicated GPO to configure for this container only.



In GPMC: Right-click the container (OU/Domain/Site) > **Link an Existing GPO...** or **Create a GPO in this domain, and Link it here...**



LINK ORDER (WITHIN A SINGLE CONTAINER)

When multiple GPOs are linked to the **SAME** container (OU/Domain/Site), the **Link Order** determines precedence for that container **ONLY**.

- Link Order 1 has the **HIGHEST** precedence.
- Lower numbers win. If settings conflict, the GPO with the lower Link Order wins.
- This does **NOT** affect other containers. Link Order is evaluated separately per container.



REORDERING LINKS IN GPMC

- In **Group Policy Management**, select the container (OU/Domain/Site).
- In the lower pane, under **Linked Group Policy Objects**, select the GPO you want to move.
- Click **Move Up** or **Move Down** to change the Link Order.
- The **Link Order** numbers update automatically.

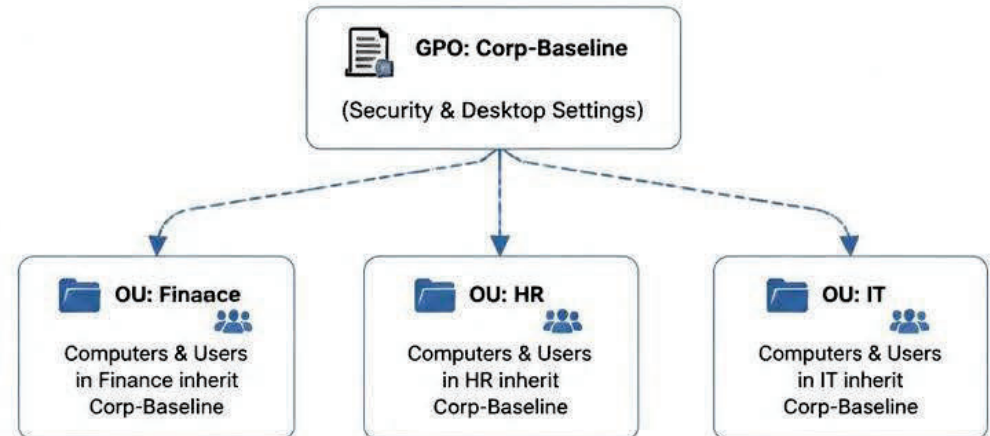
★ **TIP:** Test changes with `gpresult /r` or `rsop.msc` to confirm the resulting configuration.

Linked Group Policy Objects

Link Order	GPO
1	Sales - Workstation Restrictions
2	Corp - Workstation Baseline
3	Default Domain Policy

Move Up Move Down

ONE GPO LINKED TO MULTIPLE CONTAINERS (EXAMPLE)



MULTIPLE GPOS LINKED TO ONE OU (EXAMPLE)



1 GPO with Link Order 1 wins in case of conflict (within this OU only).



WORKED EXAMPLE: WHICH SETTING WINS?

Scenario: All three GPOs configure the same setting. "Disable Control Panel"

Link Order	GPO Name	Setting Configured	Value	Does it Win?	Reason
1	Sales - Workstation Restrictions	Disable Control Panel	Enabled	✓	Highest precedence (Link Order 1)
2	Corp - Workstation Baseline	Disable Control Panel	Not Configured	✗	Lower precedence
3	Default Domain Policy	Disable Control Panel	Disabled	✗	Lower precedence



RESULT: The setting will be **Enabled**, because the GPO with Link Order 1 wins for this OU.



SECTION 8: INHERITANCE PREVENTION (Block Inheritance)



WHAT IT DOES: Block Inheritance is set on an OU or Domain, and it **PREVENTS** GPOs linked to **ANY** parent container (parent OU, Domain, or Site) from automatically applying to that blocked container and its children.



EXAMPLE

SCENARIO:

The Domain has a GPO named "Domain Security Policy" linked at the Domain level.

The "Sales" OU has Block Inheritance turned ON.



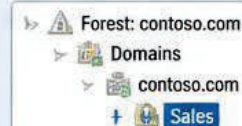
RESULT: Objects inside Sales will **NOT** receive "Domain Security Policy". They will only get GPOs linked directly to Sales or its child OUs.

HOW TO SET IT IN GPMC

- 1 In Group Policy Management Console (GPMC), right-click the OU or Domain you want to block.
- 2 Click **Block Inheritance**.
- 3 A confirmation dialog appears. Click **OK** (or choose to Remove inherited GPOs, if desired).



The OU or Domain now shows a **BLUE EXCLAMATION MARK (!)** icon in the console to indicate Block Inheritance is enabled.



⚠ CRITICAL RULE – VERY IMPORTANT! ⚠



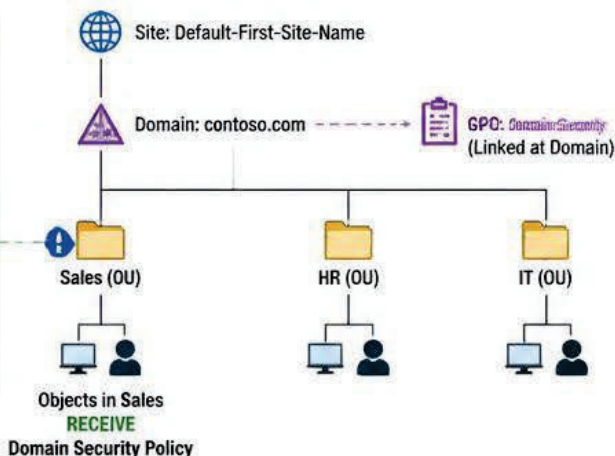
Block Inheritance does **NOT** block a GPO that is marked as **Enforced**.

ENFORCED GPOs always apply regardless of Block Inheritance.
(See Section 9: Enforced GPOs)

WITHOUT BLOCK INHERITANCE (Default Behavior)

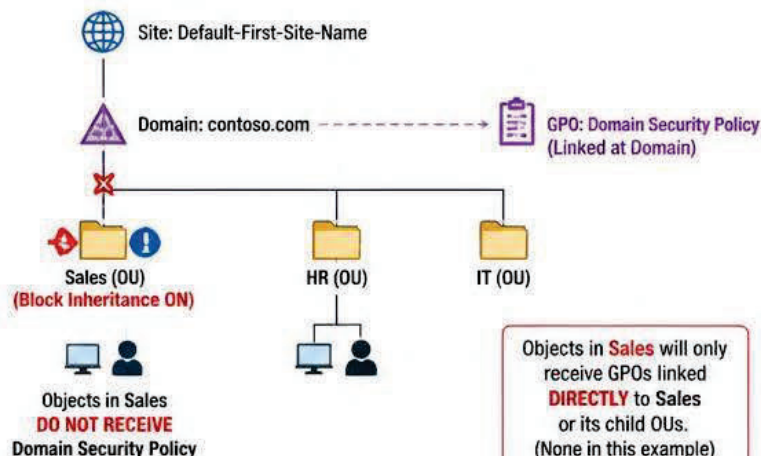
INHERITANCE CHAIN (Top to Down)

- 1 Site
- 2 Domain
- 3 Parent OU(s)
- 4 Child OU(s)/Objects



All GPOs linked to parent containers **automatically apply** to child containers and their objects.

WITH BLOCK INHERITANCE (Enabled on Sales OU)



GPOs from parent containers are **BLOCKED** from applying to Sales and its children.

Objects in Sales will only receive GPOs linked **DIRECTLY** to Sales or its child OUs. (None in this example)

LEGEND

- Site
- Domain
- Organizational Unit (OU)
- Computers / Users
- GPO
- Inheritance (Allowed)
- X- Inheritance (Blocked by Block Inheritance)
- ! Block Inheritance Enabled



KEY TAKEAWAY:

Block Inheritance breaks the automatic flow of GPOs from parent containers. Use it to isolate OUs from unwanted policies — but remember: **Enforced GPOs always win!**



SECTION 9: ENFORCING (No Override)

Enforced" (previously called "No Override" in older Windows Server versions) is a setting applied to a GPO **LINK**, not to the GPO object itself.



TWO EFFECTS OF MARKING A GPO LINK AS ENFORCED

1 OVERRIDES BLOCK INHERITANCE



The GPO's settings apply even to child OUs that have **Block Inheritance** turned ON.

2 HIGHEST POSSIBLE PRECEDENCE



Enforcing a GPO link works by moving that GPO to the very **END** of the processing order. Instead of the normal "last writer wins" rule, the Enforced GPO **ALWAYS wins** any conflicting setting, no matter where it's linked in the hierarchy.



WALKTHROUGH EXAMPLE: CONFLICT RESOLUTION STEP-BY-STEP

SCENARIO SETUP

Domain: contoso.local



GPO: Domain Security Policy
[ENFORCED]
Setting: Wallpaper = Company Logo



OU: Sales
Block Inheritance: ON



GPO: Sales Desktop Policy
Setting: Wallpaper = Sales Logo



Enforced Link (No Override)



Block Inheritance (applies to this OU and below)



Normal Link

HOW PROCESSING WORKS

A. Normal Processing Order (No Enforced Links)

GPOs are processed from top to bottom (parent to child). "Last writer wins."

1. Domain Security Policy
Wallpaper = Company Logo



2. Sales Desktop Policy
Wallpaper = Sales Logo



Final Result
Sales Logo

B. With Enforced Link (Domain Security Policy is Enforced)

The Enforced GPO is moved to the **END** of the order, after all other GPOs.

1. Sales Desktop Policy
Wallpaper = Sales Logo



2. Domain Security Policy
[ENFORCED]
Wallpaper = Company Logo



Final Result
Company Logo



ENFORCED GPO ALWAYS APPLIES LAST AND WINS, REGARDLESS OF WHERE IT IS LINKED.



**FINAL
RESULT**

The wallpaper will be "Company Logo".

Because the Enforced GPO overrides Block Inheritance completely and wins regardless.



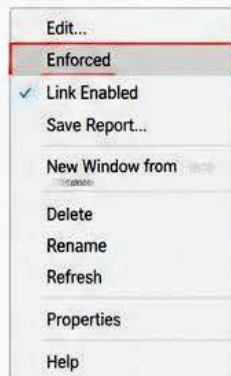
HOW TO SET A GPO LINK AS ENFORCED IN GPMC

- 1 Open Group Policy Management Console (GPMC).
- 2 Navigate to the OU or domain where the GPO is linked.
- 3 In the Linked Group Policy Objects pane, right-click the GPO link.

Linked Group Policy Objects

Link Order	GPO
1	Domain Security Policy
2	Sales Desktop Policy

- 4 Click Enforced.



A green up arrow will appear on the link.

Linked Group Policy Objects

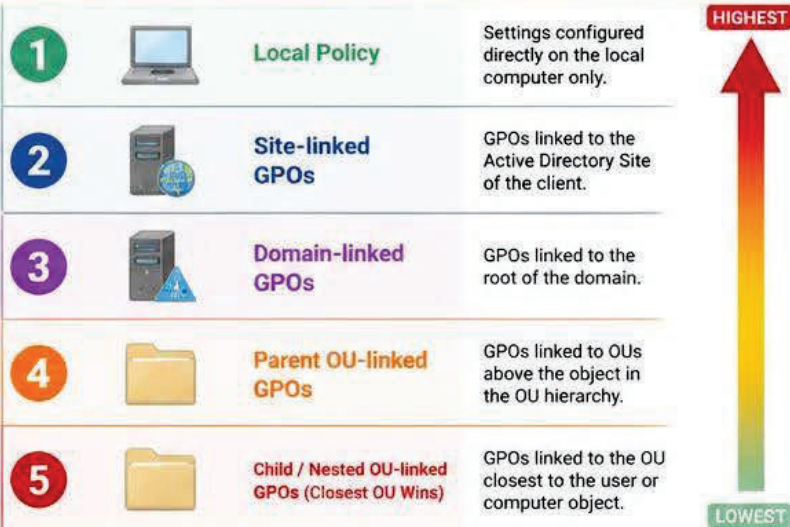
Link Order	GPO
1	Domain Security Policy
2	Sales Desktop Policy

- 5 Close the context menu. The GPO link is now Enforced.

SECTION 10: PRECEDENCE – The Complete Picture (Final Summary)

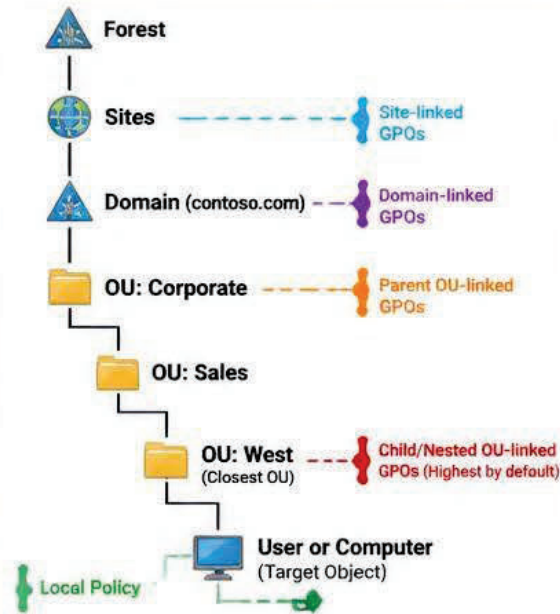
This section brings together everything from Sections 6, 7, 8, and 9 into ONE unified, final precedence model.

DEFAULT PRECEDENCE ORDER (LOWEST TO HIGHEST)



★ **Default rule:** The OU closest to the user/computer object wins by default. This is known as **LSDOU (Last Set of Denied OUs)** precedence.

WHERE THESE APPLY IN ACTIVE DIRECTORY



OVERRIDE RULE: ENFORCED

ANY GPO link marked as **Enforced** jumps to the **ABSOLUTE TOP** of precedence regardless of where it is linked in the hierarchy.



Enforced overrides ALL other GPOs (Normal or Enforced).



Enforced also overrides any **Block Inheritance** set on child containers.



ENFORCED (Always Wins)

Enforced GPO

(From ANY location: Local, Site, Domain, Parent OU, or Child OU)

Everything else comes below.

DECISION TABLE / TROUBLESHOOTING LOGIC

SCENARIO		QUESTION / SITUATION	DOES THE PARENT / OTHER GPO APPLY?	WHICH ONE WINS?	WHY / EXPLANATION
A	 Normal GPO vs Normal GPO (at different levels)	Example: Domain GPO vs Site GPO, or Parent OU GPO vs Domain GPO, etc.	 YES	The one closest to the object wins. Order: Child OU > Parent OU > Domain > Site > Local (LSDOU)	By default, Group Policy precedence is determined by proximity to the user/computer object. The closest container level has the highest precedence.
B	 Normal GPO vs Enforced GPO	Any Normal GPO compared to an Enforced GPO.	 YES	Enforced GPO always wins.	Enforced GPOs have absolute precedence over any Normal GPO, regardless of location.
C	 Block Inheritance set + Non-Enforced parent GPO	A parent OU (or higher) has a GPO (Not Enforced) and Block Inheritance is enabled on the child OU.	 NO	The parent GPO does NOT apply.	Block Inheritance prevents Non-Enforced GPOs from parent containers from applying to the child container and its objects.
D	 Block Inheritance set + Enforced parent GPO	A parent OU (or higher) has an Enforced GPO and Block Inheritance is enabled on the child OU.	 YES	The Enforced parent GPO DOES apply.	Enforced GPOs override Block Inheritance and still apply to the child container and its objects.
E	 Multiple GPOs linked to the SAME OU with different Link Orders	Two or more GPOs linked to the same OU (same container).	 YES	Link Order 1 has the highest precedence.	Within the same container, lower Link Order number (1) is processed last and therefore has the highest precedence. (1 > 2 > 3 > ...)

💡 **BEST PRACTICE:** Design GPOs with inheritance and precedence in mind. Use Enforced links sparingly and intentionally. Always test and verify with `gpresult /h report.html` or RSOP to confirm the final applied settings.



Remember: The goal is predictable, not just working, Group Policy.



QUICK REFERENCE CHEAT SHEET — Windows Server Administration & Active Directory

One-Page Guide to Core Concepts, Group Policy & Active Directory Essentials

1 AGDLP MODEL (Access Management Flow)

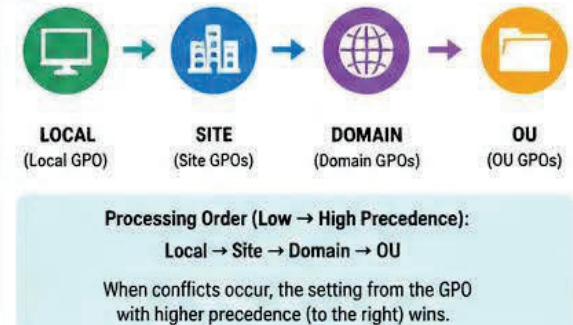


2 GROUP SCOPES QUICK REFERENCE

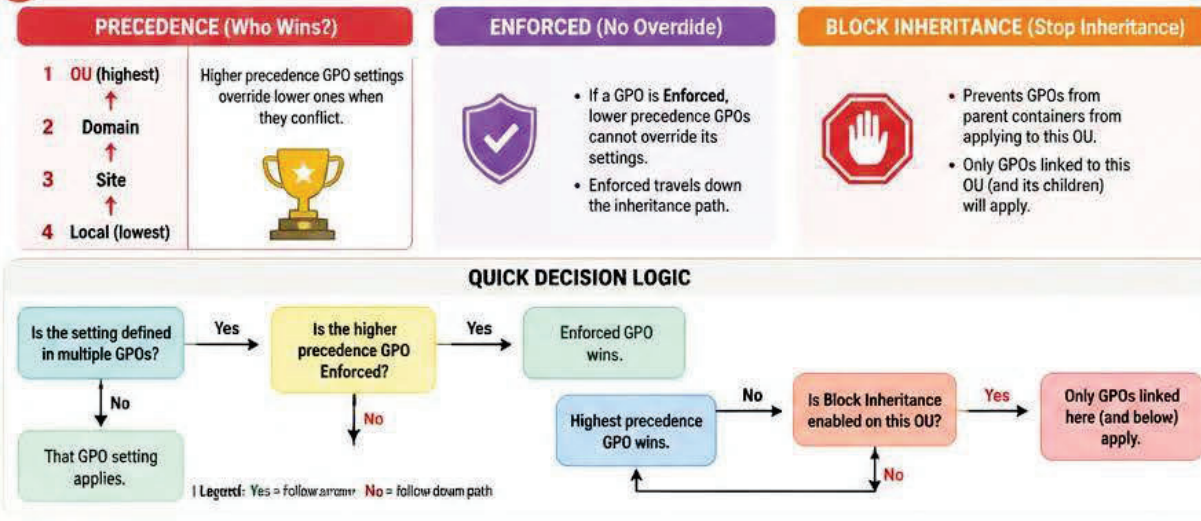
Scope	Where It Can Have Members	Where It Can Be Used	Notes / Key Use
GLOBAL	Only in its own domain	Any resource in its own domain	Best for managing users/computers. Used in AGDLP.
DOMAIN LOCAL	Users / Computers from any domain in the same forest	Only in its own domain	Used to assign permissions to resources in domain.
UNIVERSAL	Users / Computers from any domain in the forest	Any resource in any domain in the forest	Use when sharing across multiple domains.

Rule of Thumb: Use the most restrictive scope that meets your need.

3 LSDOU: GPO PROCESSING ORDER



4 PRECEDENCE, ENFORCING & BLOCK INHERITANCE — DECISION RULES

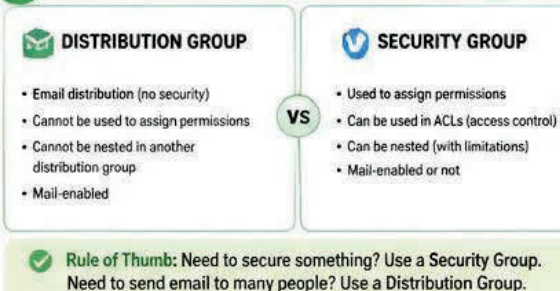


5 GPO ANATOMY

COMPUTER CONFIGURATION Applies to computers (startup)		USER CONFIGURATION Applies to users (login)	
POLICIES (Mandatory)	PREFERENCES (Non-Mandatory)	POLICIES (Mandatory)	PREFERENCES (Non-Mandatory)
• Account Policies • Local Policies • Event Log • Restricted Groups • System Services • Registry • File System • Administrative Templates (Settings)	• Drive Maps • Printers • Shortcuts • Environment • Files • Folders • INI Files • Registry	• Account Policies • Local Policies • Event Log • Restricted Groups • System Services • Registry • File System • Administrative Templates (Settings)	• Drive Maps • Printers • Shortcuts • Environment • Files • Folders • INI Files • Registry

Policies enforce settings. Preferences customize environment.

6 DISTRIBUTION GROUP vs SECURITY GROUP



7 KEY DEFINITIONS (At a Glance)

OU (Organizational Unit)	Container in AD to organize objects and apply GPOs and delegate administration.	Enforced	Prevents lower-precedence GPOs from overriding settings.
Security Group	AD group used to assign permissions to resources.	Block inheritance	Stops GPOs from parent containers from applying.
GPO (Group Policy Object)	Collection of settings that control user and computer configurations.	WMI Filter	Filters GPO application based on WMI query (e.g., OS version).
Domain Local Group	Group scope used to grant permissions to resources in the same domain.	Loopback Processing	Applies User Configuration settings from the computer's OU instead of the user's OU.
Global Group	Group scope used to manage users/computers within one domain.		
Universal Group	Group scope used to manage users/computers across the forest.		

8 EXAM / INTERVIEW QUICK TIPS

- ✓ Follow AGDLP for scalable, manageable access control.
- ✓ Use the most restrictive group scope possible.
- ✓ Remember LSDOU order: Local → Site → Domain → OU.
- ✓ Know precedence rules, and the impact of Enforced and Block Inheritance.
- ✓ Understand GPO structure: Computer vs User, Policies vs Preferences.
- ✓ Security Groups = permissions; Distribution Groups = email.